

Several benchmarks

CLEVRER



***Question*: What is the description of the glitch observed in the video?**

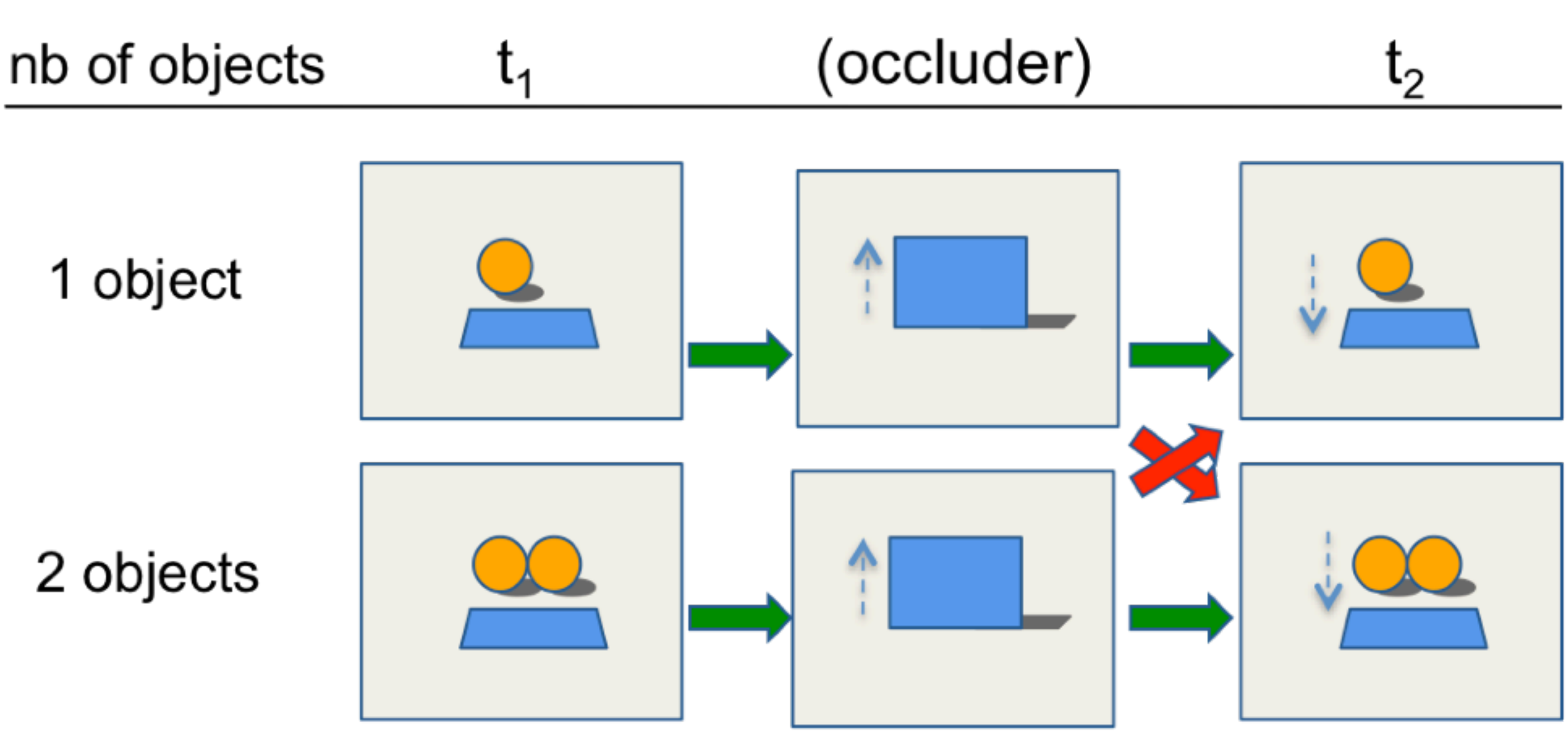
A. The character vibrated intensely and then shot straight into the sky.

B. The lamp rolling on the ground collides with a man and pushes him forward.

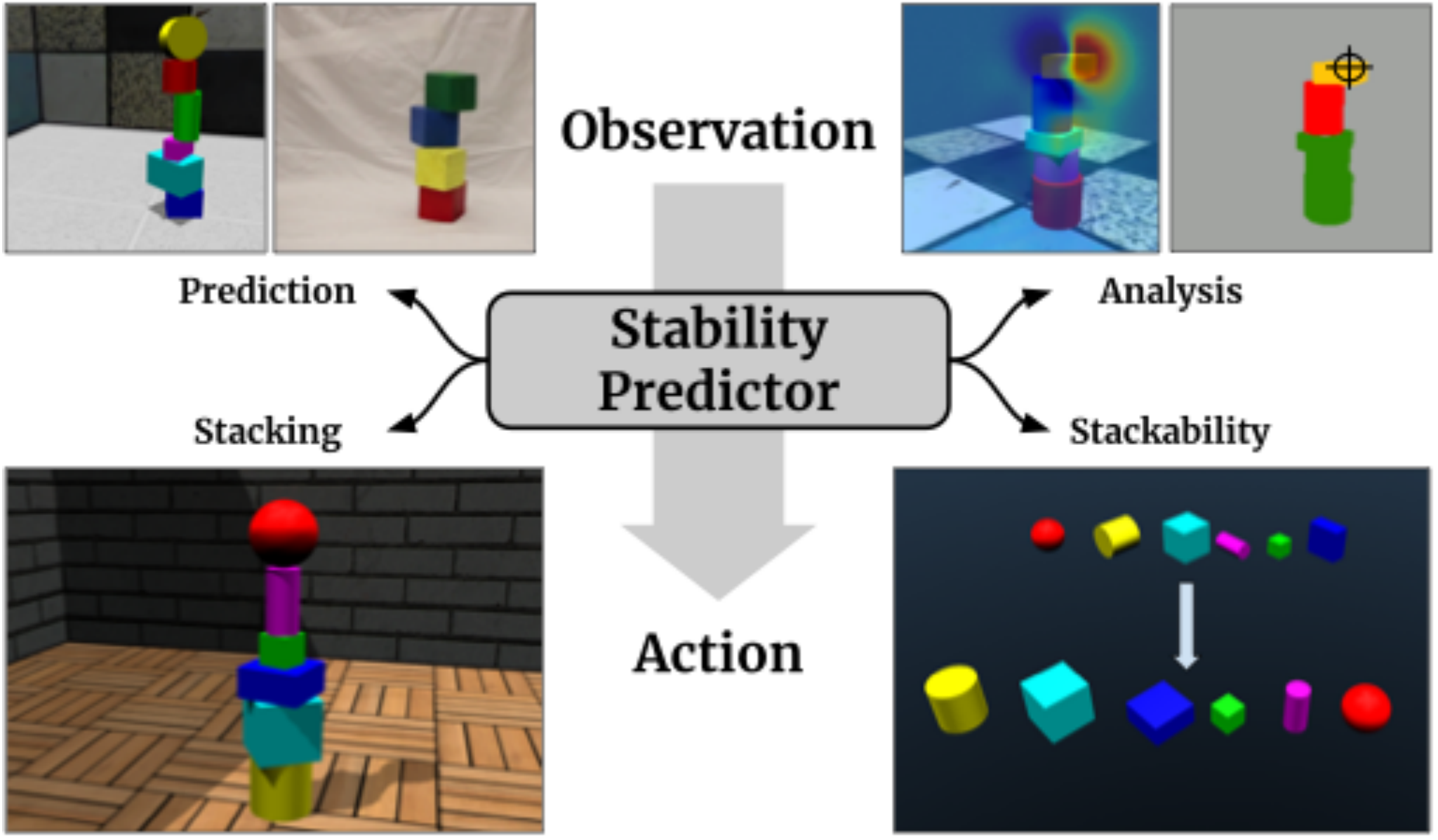
C. The man unexpectedly flew into the air in an unrealistic manner.

D. The appearance of the mountain turned into an exaggerated blue-purple color.

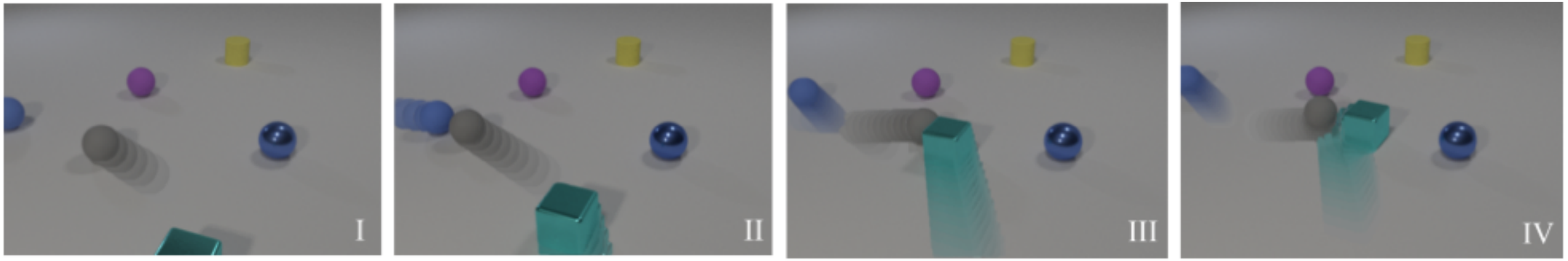
PHYSGAME


nb of objects
t_1
(occluder)
t_2
1 object
2 objects

INTPHYS


Observation
Prediction
Analysis
Stability Predictor
Stacking
Stackability
Action

SHAPESTACKS



Descriptive:

Q: *How many spheres are moving?*
A: *2*

Q: *What shape is the second object to collide with the gray object?*
A: *Cube*

Q: *Are there any collisions after the cube enters the scene?*
A: *Yes*

Explanatory:

Q: *Which of the following is responsible for the collision between the gray object and the cube?*

a) The presence of the purple object
b) The collision between the blue sphere and the gray sphere
c) The presence of the purple object
d) The presence of the blue object

A: *b), d)*

Predictive:

Q: *What will happen next?*

a) The cube and the gray object collide
b) The gray sphere collides with the purple sphere
c) The metal sphere and the cube collide
d) The gray sphere collides with the blue sphere

A: *b)*

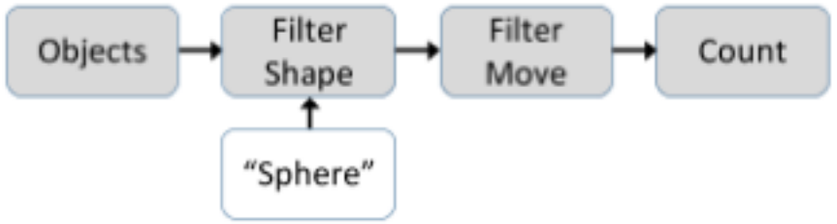
Counterfactual:

Q: *What will happen if the gray sphere is removed?*

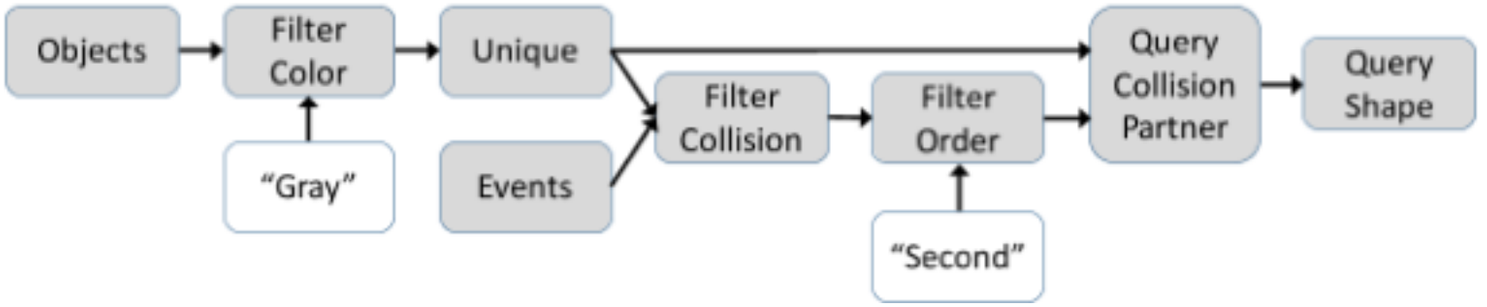
a) The blue sphere collides with the cube
b) The blue sphere and the metal sphere collide
c) The purple object collides with the cylinder
d) The cube and the metal sphere collide

A: *a), d)*

How many spheres are moving?

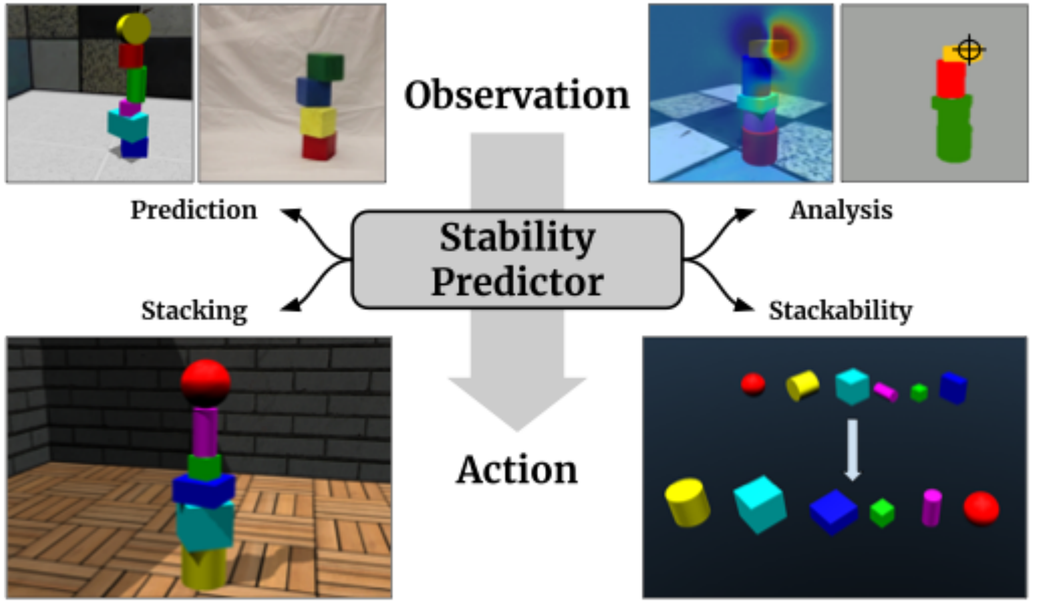


What shape is the second object to collide with the gray object?



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SHAPESTACKS



The PhyRE Dataset

of samples: